

ers were “tested on over 11,000 farm fields on 94 crops and 20+ agricultural research institutes/ universities (affiliated with the ICAR) on 43 crops” and approved by the Centre. The tested nano fertilisers were nano nitrogen (nano urea and nano DAP), nano zinc and nano copper.

The recent FCO directive capping nano fertiliser to 25% now suggests a preference for conventional fertiliser. This, then, shifts focus to the scientific rigour of IFFCO’s trials and the application of nanotech in fertiliser.

“Nano urea particles of 20–50 nanometres have a dramatically higher surface-area-to-volume ratio compared to conventional granular urea, allowing them to be absorbed directly through leaf stomata with far greater efficiency. Our laboratory and field data demonstrate that the nutrient use efficiency of nano urea is 85–90%, compared to a mere 25% for conventional urea, which is a staggering leap that any scientist or farmer should find compelling,” Patel tells *Business Today*:

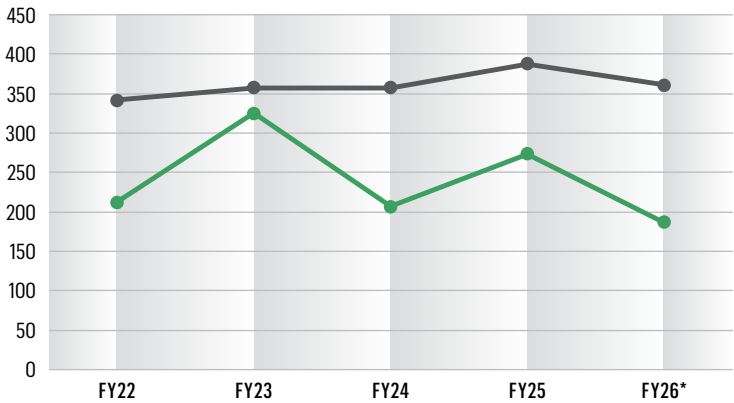
The agriculture ministry, which issues FCOs, doesn’t seem to agree with IFFCO’s claim that nano fertilisers can replace 50% of conventional fertiliser use, the basis for accruing the benefits discussed earlier.

Business Today asked Patel to share comparative data from IFFCO’s trials demonstrating the nano fertiliser’s efficiency in delivering nutrients to crops vis-à-vis conventional fertilisers. He didn’t respond.

While foliar use of conventional urea is limited to exceptional cases like drought, nano fertilisers—meant for such use—are exclusively meant for foliar use.

FEW TAKERS

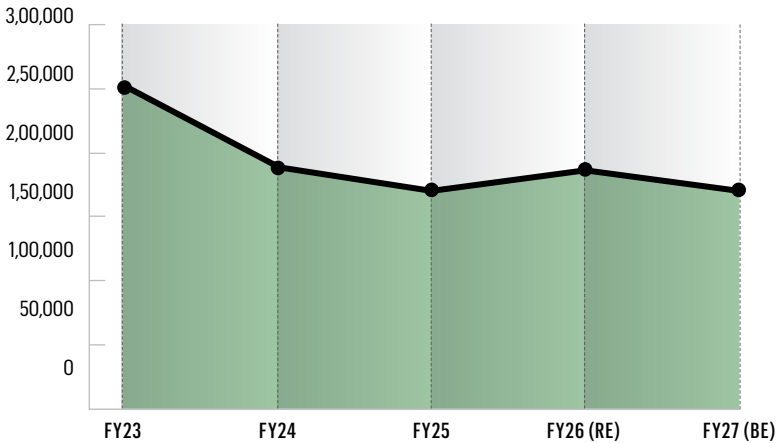
Sale of urea and nano urea since the latter was introduced in FY22



● SALE OF UREA (LAKH TONNE) ● SALE OF NANO UREA (LAKH BOTTLES)
NOTE *FY26 DATA TILL FEB 8
SOURCE RAJYA SABHA ANSWER, FEB 13, 2026

SUBSIDY OUTGO

The drive towards nano fertilisers was expected to reduce conventional fertiliser use, subsidy burden



IN ₹ CRORE **SOURCE:** BUDGET DOCUMENTS

CAPACITY CHALLENGES

IFFCO claims its trials showed nano fertilisers increasing crop yield by 3-8%. However, trials by the Punjab Agricultural University (PAU), also sponsored by IFFCO and published in December

2024, challenged this finding and led the agriculture ministry to stop forced sales and order retrials.

The PAU tested nano urea on wheat and paddy in 2020-21 and 2021-22. It followed the IFFCO